

Week 3: BERTweet Integration and ML Classification

Project: RESQAI — AI-Assisted Humanitarian Disaster Message Classification

ML Service

The third week focused on integrating machine learning. A separate Python FastAPI service was created to handle inference. Its services include BERTweet classification and Gemini integration, with the main application exposing a classification endpoint.

BERTweet Classification

BERTweet was selected as the primary classification model. It returns a category and confidence score. The system supports ten categories: `caution_and_advice`; `displaced_people_and_evacuations`; `infrastructure_and_utility_damage`; `injured_or_dead_people`; `missing_or_found_people`; `not_humanitarian`; `other_relevant_information`; `requests_or_urgent_needs`; `rescue_volunteering_or_donation_effort`; and `sympathy_and_support`.

Confidence-Based Routing

A confidence threshold of 0.70 was established. If confidence is at least 0.70 and the predicted category is not `not_humanitarian`, the BERTweet result is used. Otherwise, the message is sent to Gemini for additional analysis.

Example

For the message “We urgently need water and medical supplies,” BERTweet may return category `requests_or_urgent_needs` with confidence 0.92. Since the confidence exceeds 0.70 and the category is not `not_humanitarian`, the system uses the BERTweet result without calling Gemini.

Outcome

By the end of Week 3, the primary ML classification pipeline was integrated with the backend, allowing submitted disaster messages to be automatically classified.